

Health Wearable Device

PCB Design — Component Analysis Report

Band01_Rev1 | Project Work — Task 1

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Date: 18 May 2026

1. Introduction

This report presents a detailed analysis of the PCB design for the Health Wearable Device (Band01, Revision 1), designed by Ricardo K and Gabriel P at Hochschule Hamm-Lippstadt. The device is a wrist-worn health monitoring smartband intended to continuously collect physiological data from a patient and transmit it wirelessly via Bluetooth Low Energy (BLE).

The report describes each major component used in the design, its function within the system, and how it contributes to the overall operation of the device. The goal is to establish a thorough understanding of the hardware before proceeding to firmware analysis and functional testing.

2. System Overview

The Health Wearable Device is built around a Texas Instruments CC2640R2F Bluetooth Low Energy microcontroller, which serves as the central processing and wireless communication unit. The device measures three physiological parameters:

- Heart rate and blood oxygen saturation (SpO2) — MAX30102 optical sensor
- Body temperature — MAX30205 digital temperature sensor
- Motion and acceleration — LIS331DL 3-axis MEMS accelerometer

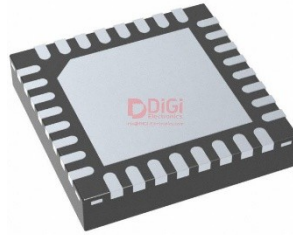
All three sensors communicate with the microcontroller via a shared I2C bus. Power is supplied from a battery (VBAT) and regulated to 3.3V and 1.8V by two LP5907 LDO regulators. Two push buttons provide user interaction. A PCB trace antenna enables wireless BLE communication. An 18-pin expansion header exposes GPIO, power, and JTAG programming lines for development use.

The PCB measures 34mm × 40mm with a VDD5 power plane and a 3V3 power plane clearly separated in the layout, and the MAX30102 sensor is positioned at the bottom for direct skin contact during wrist-worn use.

3. Component Analysis

3.1 Microcontroller — CC2640R2FRSMR (U1)

The CC2640R2F is a 32-bit ARM Cortex-M3 wireless microcontroller from Texas Instruments, specifically designed for Bluetooth Low Energy (BLE) 4.2 and 5.0 applications. It is the central unit of the wearable device, responsible for:



- Reading sensor data via I2C and SPI interfaces
- Processing and storing measurement data
- Transmitting data wirelessly via BLE to a smartphone or gateway
- Managing power states to extend battery life
- Handling user input from push buttons

The CC2640R2F is housed in an RSM0032B (32-pin QFN) package. In the schematic it connects to all major peripherals through its DIO (digital I/O) pins. DIO_5 and DIO_6 are used as I2C SCL and SDA lines shared by the sensors. DIO_0 and DIO_1 are connected to the two push buttons. DIO_8 serves as the interrupt line from the MAX30102 heart rate sensor. JTAG pins (DIO_3, DIO_4, JTAG_TCK, JTAG_TMS) are exposed on the P1 header for firmware programming and debugging using Code Composer Studio.

3.2 Heart Rate and SpO2 Sensor — MAX30102EFD+ (UIC1)



The MAX30102 is a high-sensitivity pulse oximetry and heart rate sensor from Maxim Integrated. It integrates two LEDs (red and infrared), a photodetector, and signal processing circuitry into a single compact package (PSON14). The sensor works by shining light through the skin and measuring the amount of light reflected or absorbed, which varies with blood flow and oxygen saturation.

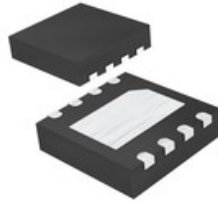
- Measures heart rate (BPM) and blood oxygen saturation (SpO2 %)
- Communicates with the CC2640R2F via I2C (SCL on DIO_5, SDA on DIO_6)
- Generates an interrupt on DIO_8 when new data is available
- Powered at 1.8V by the LP5907MFX-1.8/NOPB regulator
- Red LED drive current controlled via R_DRV pin; IR LED via IR_DRV pin

The schematic shows decoupling capacitors C108 (0.1uF) and C109 (10uF) placed close to the sensor for power supply stability, which is critical for accurate optical measurements.

3.3 Body Temperature Sensor — MAX30205MTA+T (U104)

Manufacturer: Analog Devices (formerly Maxim Integrated) | Part: MAX30205MTA+T |
Package: TDFN3X3 (9-pin) | Supplier: Analog Devices

The MAX30205 is a clinical-accuracy digital temperature sensor capable of measuring human body temperature in the range of 0°C to 45°C with a resolution of 0.00390625°C (16-bit) and accuracy of $\pm 0.1^\circ\text{C}$ — meeting clinical thermometry standards. It communicates via I2C and shares the same bus as the MAX30102 and LIS331DL.



- Measures body temperature with $\pm 0.1^\circ\text{C}$ clinical accuracy and 16-bit resolution
- I2C shared bus: SCL on DIO_5, SDA on DIO_6 | I2C address: 0x90 (A0/A1/A2 = GND)
- OS pin enables autonomous temperature threshold alerting in hardware
- 3.3V supply; pull-up R102 (10k) on address lines; decoupling C112 (0.1uF)
- Address pins A0/A1/A2 allow up to 8 devices on one I2C bus

3.4 3-Axis Accelerometer — LIS331DLHTR (U103)

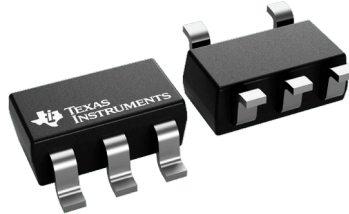


The LIS331DL is a 3-axis MEMS (Micro-Electro-Mechanical Systems) capacitive accelerometer from STMicroelectronics. It measures linear acceleration along three orthogonal axes (X, Y, Z) with a selectable range of $\pm 2g$, $\pm 4g$, or $\pm 8g$, and an output data rate configurable from 50Hz to 1kHz. It is used in the wearable to monitor patient movement, activity level, posture, and detect potential fall events.

- 3-axis acceleration measurement with configurable range ($\pm 2g/\pm 4g/\pm 8g$) and data rate
- Configurable measurement range and output data rate
- Two interrupt outputs (INT_1 on DIO_9, INT_2 on DIO_2) for motion detection events
- SDA/SDI/SDO pin connected to DIO_6, SCL/SPC to DIO_5
- Housed in a 16-pin LGA package
- 3.3V supply; decoupling C105 (0.1uF) and C106 (10uF) placed near the IC

3.5 Voltage Regulators — LP5907MFX-3.3 and LP5907MFX-1.8 (U101, U102)

The LP5907 is an ultra-low-noise LDO (Low Dropout) linear voltage regulator from Texas Instruments, available in two versions used in this design. U101 provides a regulated 3.3V supply (NL3V3) for the CC2640R2F and most digital circuitry, while U102 provides a regulated 1.8V supply (NL1V8) specifically for the MAX30102 sensor.



- U101 (LP5907MFX-3.3): outputs 3.3V from VBAT battery input
- U102 (LP5907MFX-1.8): outputs 1.8V for the MAX30102 sensor
- Both regulators feature extremely low noise, important for sensitive analog measurements
- Enable pins allow the CC2640R2F to power-gate sensors for energy saving
- Housed in SOT-23-5 packages (5-pin)

Multiple decoupling capacitors (C101, C103, C104, C107 at 1uF each) are placed at the inputs and outputs of the regulators to suppress noise and ensure voltage stability.

3.6 Push Buttons — EVQP7J01P (SW101, SW102)



Two ultra-miniature SMD tactile push buttons from Panasonic provide the only direct user input to the wearable device. They are designed for portable electronics applications with a low profile and very small footprint suitable for the compact 34×40mm PCB. Each button is connected to a DIO pin on the CC2640R2F through a 100 Ohm series resistor for current limiting and ESD protection.

- SW101 → DIO_0 via R110 (100Ω); SW102 → DIO_1 via R111 (100Ω)
- R100 and R120 (10k) pull-up to 3.3V — logic HIGH when not pressed
- Series resistors provide current limiting and ESD protection on GPIO lines
- Suitable for wearable applications due to compact SMD footprint

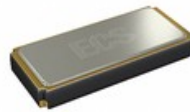
3.7 24MHz Crystal Oscillator — NX2016SA (Y1)



Y1 is the 24MHz main crystal oscillator required for BLE radio operation in the CC2640R2F. It is connected to X24M_P (pin 31) and X24M_N (pin 30) of the microcontroller and drives both the main system CPU clock and the integrated RF radio core. Without this crystal, the BLE radio cannot operate, as it requires a precise 24MHz reference for frequency synthesis at 2.4GHz.

- Required for BLE 2.4GHz radio core operation and main CPU clock — cannot be omitted
- Connected to X24M_P (pin 31) and X24M_N (pin 30) of the CC2640R2F
- C22/C23 (12pF) footprints present but unpopulated (DNM) — internal caps sufficient

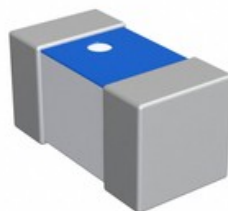
3.8 32kHz Crystal Oscillator — ECS-.327-7-34B-TR (Y2)



Y2 is the 32.768kHz low-frequency crystal oscillator for real-time clock and deep sleep wake-up timing. It is connected to X32K_Q1 (pin 5) and X32K_Q2 (pin 6) of the CC2640R2F. This crystal enables the microcontroller to maintain accurate timekeeping and BLE connection interval scheduling while operating in deep sleep with a current consumption below 1 μ A — a critical requirement for extending battery life in a wearable device.

- Provides RTC timing and BLE advertising interval accuracy during deep sleep
- Connected to X32K_Q1 (pin 5) and X32K_Q2 (pin 6) of the CC2640R2F
- Load capacitors C1 and C2 (12pF, Murata GRM0335C1H120JA01D) are populated

3.9 RF Matching Inductors — L12 (2nH) and L21 (15nH)



L12 and L21 are high-frequency chip inductors from Murata's LQP03T series, housed in the extremely compact 0201 package. They form the RF impedance matching network between the CC2640R2F's differential RF output pins (RF_P and RF_N) and the PCB trace antenna,

together with capacitor C14 (12pF). At 2.4GHz, even a 2nH inductor presents significant reactance (approximately 30 Ohms), so precise component values are critical.

- L21 (15nH) — primary RF impedance transformation from chip RF port to 50 Ohm
- L12 (2nH) — fine-tuning series element for precise 2.4GHz impedance match
- C14 (12pF, Murata GRM0335C1E120JA01D, 0201) completes the LC matching network

3.10 Power Inductor — LBMF1608T100K / L1 (10uH)



L1 is a 10uH power inductor used in the CC2640R2F's internal DC-DC switching converter circuit. The CC2640R2F contains an integrated DC-DC boost/buck converter that generates the VDDS internal supply voltage from the DCDC_SW pin (pin 18). L1, together with decoupling capacitors C8 (10uF) and C5 (10uF), forms the external LC filter required for this internal converter.

- 10uH inductance in 0603 package — external LC filter for CC2640R2F DC-DC converter
- DCDC_SW (pin 18) → L1 → VDDS supply rail; C5/C8 (10uF) as output capacitors

Improves power efficiency versus pure LDO regulation, extending battery life

3.11 Ferrite Bead — BLM18HE152SN1D / FL1



FL1 is a chip ferrite bead placed in series on the 3.3V external power supply line entering the PCB. Ferrite beads act as frequency-dependent resistors — they have very low impedance at DC, allowing current to pass freely, but present high impedance at radio frequencies, effectively blocking high-frequency noise. The BLM18HE152SN1D has an impedance of 1500 Ohms at 100MHz.

- Impedance: 1500Ω at 100MHz — effective board-level EMI and RF noise suppression
- Placed in series between the 3V3 external supply input and the internal power planes
- Protects analogue sensor circuits and RF section from supply-borne interference

3.12 Expansion Header — HTSW-109-07-G-D / P1



P1 is an 18-pin (9×2) through-hole pin header from Samtec that provides full access to the CC2640R2F's GPIO, power, and programming interfaces. It is the primary connection point between the PCB and external development tools, including a TI XDS110 JTAG debug probe for Code Composer Studio, as well as the Raspberry Pi used during the prototype development phase of this project.

- Exposes DIO_0–DIO_9, JTAG_TCK/TMS/TDI/TDO, nRESET, VBAT, 3V3, and GND
- Primary interface for TI Code Composer Studio JTAG programming and debugging
- Connected to Raspberry Pi during prototype phase for serial sensor data readout

3.13 PCB Trace Antenna — A (IF_Antenna)



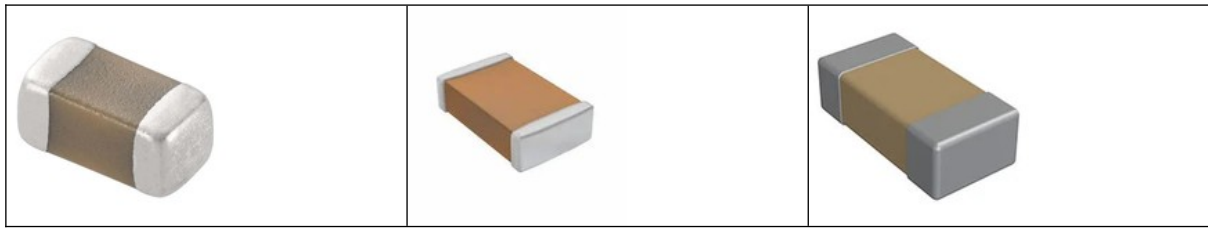
The antenna is an inverted-F style copper trace pattern on the top layer of the PCB — no separately purchased component is required. Its geometry is defined per Texas Instruments Application Note AN043 (SWRA117D.pdf), which provides reference antenna designs for the CC2640R2F and related BLE devices. The transmission line connecting the RF matching network to the antenna is sized for 50-Ohm characteristic impedance per JLCPCB fabrication specifications.

- 2.4GHz BLE antenna — copper trace on PCB top layer, no separate part required
- 50-Ohm transmission line; designed per TI AN043 (SWRA117D.pdf)
- Connected to RF_P/RF_N (pins 1/2) of CC2640R2F via L12, L21, and C14

4. Passive Components

The remaining components not individually described in Section 3 are the decoupling capacitors and signal conditioning resistors distributed throughout the PCB. Although individually small, these passive components are essential for power supply stability, I2C bus signal integrity, crystal oscillator operation, RF impedance matching, and GPIO protection. Their confirmed manufacturer part numbers are taken directly from the Aisler BOM of March 2026.

4.1 Capacitors

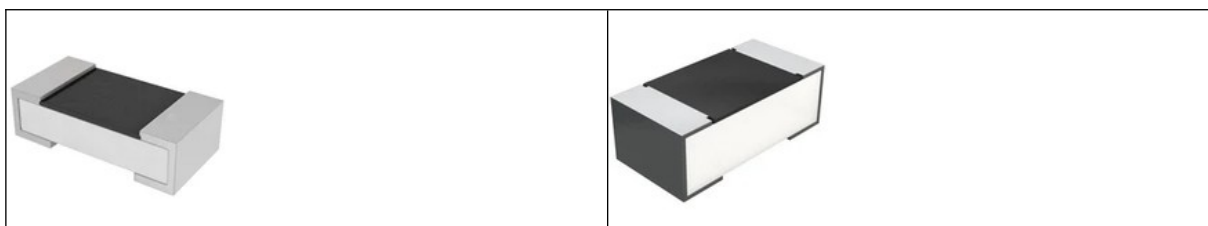


Left to right: Murata GCM series MLCC, Samsung CL series MLCC, KEMET C0402 series MLCC. All are surface-mount ceramic capacitors in 0201 or 0402 packages.

The design uses a large number of Multi-Layer Ceramic Capacitors (MLCCs) from Murata, Samsung, and KEMET distributed throughout the board. Different capacitor values serve distinct purposes: small capacitors (1.2pF, 12pF) are used in the crystal and RF matching circuits where precise capacitance is critical; mid-range capacitors (0.1uF) provide high-frequency decoupling placed directly beside each IC; and large capacitors (1uF, 4.7uF, 10uF) provide bulk energy storage for the voltage regulators and sensor supply rails.

- C1, C2 — GRM0335C1H120JA01D (Murata, 12pF, 0201) — 32kHz crystal Y2 load capacitors, setting the correct resonant frequency
- C3, C4, C6, C9, C10, C20 — GRM03C80J104KE15D (Murata, 0.1uF, 0201) — general power supply decoupling capacitors placed throughout the board beside the CC2640R2F supply pins
- C5, C8, C106, C109 — GRM155C80J106ME11D (Murata, 10uF, 0402) — bulk decoupling on the DC-DC converter output and sensor supply rails, absorbing larger current transients
- C19, C101, C103, C104, C107 — CL05A105MQ5NNNC (Samsung, 1uF, 0402) — LP5907 regulator input and output filter capacitors ensuring supply stability under varying load
- C105, C108, C110, C112 — C0402C104M9PACTU (KEMET, 0.1uF, 0402) — high-frequency decoupling capacitors placed directly beside LIS331DL, MAX30102, and MAX30205 to suppress switching noise at the sensor supply pins
- C111 — GRM155R60J475ME47D (Murata, 4.7uF, 0402) — low-frequency supply filter for the MAX30102 analogue section, reducing voltage ripple that would otherwise appear as noise in optical measurements
- C22, C23 — GRM0335C1H120JA01D (Murata, 12pF, 0201) — marked DNM (Do Not Mount); optional 24MHz crystal load capacitors that are not required as the CC2640R2F has sufficient internal capacitance

4.2 Resistors



Left: Yageo RC0402 series thick-film resistors. Right: Vishay CRCW0402 series precision resistors. All in 0402 SMD packages.

The resistors in this design serve three distinct functions: pull-up resistors establish defined logic levels on I2C lines and GPIO inputs; series resistors protect GPIO pins from excessive current and ESD events; and zero-ohm resistors act as configurable PCB jumpers that route signals without requiring physical wire links. All resistors are in the compact 0402 package (1.0mm × 0.5mm), consistent with the miniaturised wearable form factor.

- R1 — RC0402JR-1010KL (Yageo, 100k, 0402) — pull-up resistor on the nRESET line to 3.3V, ensuring the CC2640R2F starts in a defined state at power-on and is not held in reset by floating input conditions
- R15, R16 — RC0402JR-070RL (Yageo, 0Ω, 0402) — zero-ohm jumpers on the JTAG_TDO and JTAG_TDI lines used to configure the routing of these debug signals on the PCB
- R100, R102, R103, R104, R105, R120 — RC0402FR-0710KL (Yageo, 10k, 0402) — I2C bus pull-up resistors (R103 on SCL, R104 on SDA) that hold the bus lines HIGH when idle, along with pull-ups on button lines (R100, R120) and sensor address/interrupt lines (R102, R105)
- R110, R111 — CRCW0402100RJNED (Vishay, 100Ω, 0402) — series protection resistors on SW101 (DIO_0) and SW102 (DIO_1), limiting inrush current and providing a degree of ESD protection for the CC2640R2F GPIO pins
- R121 — RC0402JR-070RL (Yageo, 0Ω, 0402) — zero-ohm configuration jumper on the DIO_9 signal path used for accelerometer interrupt routing
- R17, R18 — DNM (Do Not Mount) — optional configuration resistor pads left unpopulated in Revision 1 of the board, providing future design flexibility without requiring a PCB revision

5. Communication Architecture

All three sensors (MAX30102, MAX30205, LIS331DL) share a single I2C bus connected to DIO_5 (SCL) and DIO_6 (SDA) of the CC2640R2F. Pull-up resistors R103 and R104 (10k each) keep the bus lines high when idle. Each sensor has a unique I2C address, allowing the microcontroller to address them individually on the same two-wire bus.

The CC2640R2F processes the collected sensor data internally and transmits it wirelessly via Bluetooth Low Energy (BLE) to a paired device such as a smartphone or Raspberry Pi gateway. The JTAG interface on header P1 connects to Texas Instruments Code Composer Studio for firmware development and debugging.

6. Power Architecture

The device is powered from a battery (VBAT). The CC2640R2F contains an internal DC-DC converter (DCDC_SW) that generates the VDDS supply for the digital core, assisted by external inductor L1 (10uH) and capacitors. The internal VDDR supply powers the RF circuitry.

Two external LP5907 LDO regulators provide clean regulated voltages: U101 generates 3.3V (NL3V3) for the accelerometer, buttons, and other 3.3V logic, while U102 generates 1.8V (NL1V8) for the MAX30102 sensor which requires a lower supply voltage. The enable pins of both regulators are controlled by the CC2640R2F, allowing sensors to be powered down completely during sleep modes to conserve battery.

7. Unassigned and DNM Components

The Aisler BOM (March 2026) lists 5 components as unassigned or not yet sourced. None of these affect basic device functionality:

- A (IF_Antenna) — the PCB trace antenna is a copper pattern etched into the PCB itself and requires no purchased component. It is designed per TI AN043 (SWRA117D.pdf) with a 50-Ohm line width per JLCPCB specification.
- C22, C23 — optional load capacitors for the 24MHz crystal Y1. Footprints are present but marked DNM in the schematic, as the CC2640R2F has sufficient internal capacitance for the chosen NDK crystal.
- R17, R18 — marked DNM (Do Not Mount) in both the schematic and Aisler BOM. These 0402 pads are reserved for optional signal routing configuration and remain unpopulated in Revision 1.

8. Conclusion

The Health Wearable Device (Band01_Rev1) is a professionally designed, compact BLE-based health monitoring platform. The Aisler BOM of March 2026 confirms 52 assigned components sourced from Texas Instruments, Analog Devices, STMicroelectronics, Murata, Panasonic, Yageo, Vishay, Samtec, Taiyo Yuden, Samsung, KEMET, NDK, and ECS.

The three sensor ICs — MAX30102 (heart rate/SpO2), MAX30205 (temperature), and LIS331DL (acceleration) — all share a single I2C bus and collectively cover the three vital signs targeted by the project. The CC2640R2F BLE microcontroller orchestrates all sensing, processing, and wireless transmission. The dual LP5907 regulator design with firmware-controlled enable pins enables aggressive power management essential for a wearable battery-powered device. The RF matching network formed by L12 (2nH), L21 (15nH), and C14 (12pF) ensures efficient BLE signal transmission via the PCB trace antenna.

With this hardware foundation thoroughly understood, the next step in the project is to obtain access to the firmware repository, study the code structure in Texas Instruments Code Composer Studio, and prepare the second report describing the firmware architecture and its interaction with each of the hardware components documented in this report.